LITERATURE SURVEY

Project Title

EXPLAINABLE CONCRETE MIX OPTIMIZATION USING ML

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INTRODUCTION

Concrete is one of the most widely used construction materials in the world due to its strength, durability, and cost-effectiveness. The compressive strength of concrete is a critical parameter that determines its structural performance and durability. Traditionally, compressive strength is determined through laboratory experiments, which require significant time, labor, and financial resources. These conventional methods also delay construction activities because concrete specimens must be cured for several days before testing.

Recent developments in Artificial Intelligence (AI) and Machine Learning (ML) have provided efficient alternatives for predicting concrete compressive strength using mix design parameters. Machine learning algorithms can analyze complex relationships between concrete ingredients such as cement, water, fly ash, blast furnace slag, coarse aggregate, fine aggregate, superplasticizer, and curing age. These models produce accurate predictions while reducing testing time and experimental costs.

Several machine learning techniques, including Decision Tree, Random Forest, Support Vector Regression (SVR), Artificial Neural Networks (ANN), XGBoost, LightGBM, CatBoost, Gradient Boosting, and Bagging Regressor, have been successfully applied for concrete strength prediction. Among these algorithms, ensemble learning methods consistently achieve higher prediction accuracy due to their ability to model nonlinear relationships and minimize prediction errors.

Recently, Explainable Artificial Intelligence (XAI) has become an emerging research area that enhances the transparency of machine learning models. Techniques such as SHAP (Shapley Additive Explanations) enable engineers to understand how individual input parameters influence prediction results. Explainable AI increases the reliability of machine learning models and supports better engineering decision-making.

This literature survey reviews recent research contributions related to machine learning-based concrete compressive strength prediction and optimization. The selected studies are analyzed to understand the algorithms used, their performance, advantages, and limitations. The survey also provides insights into the current research trends and motivates the development of an explainable machine learning framework for optimized concrete mix design.

OBJECTIVES OF LITERATURE SURVEY

The objectives of this literature survey are:

- To study recent research on machine learning techniques for concrete compressive strength prediction.
- To understand various algorithms used for predicting concrete strength.
- To analyze the advantages and limitations of existing prediction models.
- To compare the performance of different machine learning algorithms.
- To understand the role of Explainable Artificial Intelligence (XAI) in improving model transparency.
- To identify existing challenges in concrete mix optimization and prediction.
- To provide the background required for the proposed project.

DETAILED LITERATURE SURVEY (PAPER 1–PAPER 8)

Paper 1: **Early Prediction of Characteristic Compressive Strength of Concrete Based on Mix Proportions Using Modified Dimensional Analysis (2024)**

This study proposed a Modified Dimensional Analysis (MDA) approach for predicting the compressive strength of concrete using mix proportions. The proposed method was compared with traditional machine learning models and demonstrated high prediction accuracy while requiring fewer computational resources. The model effectively predicted concrete strength using a limited number of input parameters, making it suitable for early-stage concrete quality assessment.

Paper 2: **Predicting the Compressive Strength of Concrete by Using Machine Learning Techniques (2024)**

This research evaluated several machine learning algorithms, including Support Vector Regression, Decision Tree, Random Forest, Gradient Boosting, AdaBoost, Bagging Regressor, and XGBoost. The study concluded that ensemble learning algorithms, particularly XGBoost and Random Forest, achieved superior prediction accuracy compared to traditional regression methods. The research demonstrated the effectiveness of machine learning in reducing laboratory testing requirements.

Paper 3: **Predicting Compressive Strength of Concrete Incorporating Fly Ash, Blast Furnace Slag, and Superplasticizer Using Machine Learning Techniques (2024)**

The researchers investigated the prediction of compressive strength for sustainable concrete mixtures containing supplementary cementitious materials. Multiple machine learning algorithms such as Linear Regression, Random Forest, Decision Tree, LightGBM, CatBoost, AdaBoost, and XGBoost were evaluated. CatBoost and XGBoost produced the highest prediction accuracy with minimum prediction errors, proving their effectiveness for sustainable concrete applications.

Paper 4: **Machine Learning Models for Concrete Strength Prediction (2025)**

This study compared different machine learning algorithms for predicting concrete compressive strength using various mix design parameters. Models such as Support Vector Regression, Random Forest, Artificial Neural Networks, and XGBoost were analyzed. The study reported that XGBoost provided the highest prediction performance while effectively capturing nonlinear relationships among concrete ingredients.

Paper 5: **Machine Learning Approaches for Forecasting Compressive Strength of High-Strength Concrete (2025)**

This research focused on predicting the compressive strength of high-strength concrete using Artificial Neural Networks, Random Forest, and XGBoost. The optimized machine learning models significantly reduced prediction errors and provided accurate strength estimation. The study demonstrated that machine learning techniques can efficiently replace conventional laboratory testing for strength prediction.

Paper 6: **Optimized Machine Learning Algorithms with SHAP Analysis for Predicting Compressive Strength in High-Performance Concrete (2025)**

The study introduced Explainable Artificial Intelligence (XAI) using SHAP to improve the interpretability of machine learning predictions. Multiple machine learning models including Random Forest, XGBoost, Artificial Neural Networks, Support Vector Regression, and Gradient Boosting were evaluated. XGBoost achieved the highest prediction accuracy, while SHAP analysis successfully explained the contribution of each input variable.

Paper 7: **Concrete Compressive Strength Classification Using Hybrid Machine Learning Models and Interactive GUI (2025)**

This research proposed a hybrid machine learning framework integrating Random Forest, LightGBM, CatBoost, AdaBoost, and XGBoost. Bayesian optimization was employed to improve model performance. LightGBM achieved the highest prediction accuracy, and an interactive graphical user interface was developed to assist engineers in practical applications.

Paper 8: **Machine Learning-Based Prediction of Concrete Compressive Strength (2024)**

The authors compared Artificial Neural Networks, Random Forest, Gradient Boosting, and Support Vector Regression for predicting concrete compressive strength. The study concluded that Artificial Neural Networks produced the most accurate predictions for large datasets, while Random Forest offered stable performance with reduced overfitting. The importance of feature selection in improving prediction accuracy was also emphasized.

SUMMARY

The reviewed studies show that XGBoost, Random Forest, LightGBM, CatBoost, and ANN are effective for concrete compressive strength prediction. Explainable AI techniques such as SHAP improve transparency and support engineering decision-making.